Calibration Split Instability in Mixed-Precision LLM Quantization

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Paper ID withheld for double-blind review

July 2, 2026

Abstract

Mixed-precision post-training quantization (PTQ) often relies on a small calibration set to rank modules by sensitivity and allocate scarce high-bit precision. We reformulate this practice as a constrained calibration-driven optimization problem rather than a one-shot ranking heuristic. The central failure mode is calibration split instability, where equally plausible small prompt samples induce materially different module-sensitivity rankings. We introduce Calibration Split Instability (CSI), an auditable diagnostic and soft constraint for calibration-driven allocation. CSI evaluates seed-pair agreement using rank correlation, top-k overlap, positive-set overlap, bootstrap trend tests, and permutation-null tests before a calibration-derived allocation is promoted to a method claim. Across Qwen2.5 same-prompt-pool artifacts, increasing calibration size improves stability: for Qwen2.5-1.5B, mean Spearman rises from 0.1410 at $n = 2$ to 0.4918 at $n = 8$, with positive bootstrap gain intervals and Holm-adjusted permutation $p \leq 0.00015$. A stricter second-pool SmolLM2-360M replication shows the same trend from $n = 4$ to $n = 16$, with mean Spearman improving from 0.3133 to 0.6959 and Holm-adjusted permutation $p = 0.0005999$. We further confront native PTQ baselines on full Qwen2.5-1.5B downstream benchmarks: on MMLU (14,042 examples), FP16/AutoAWQ/GPTQModel reach 0.5864/0.5648/0.5362, and on GSM8K (1,319 examples), 0.0811/0.0788/0.0728. We also add local GPU fake-quant allocation smoke tests in which CSI/consensus 4-to-8 allocation improves PPL over uniform INT4 on Qwen2.5-0.5B slices and a Qwen2.5-1.5B single-slice feasibility run. The resulting empirical organization is a claim-boundary audit: which stability regimes are already measured, which allocation failure modes are separated, and which same-budget downstream rows remain required before promoting the audit into a method-superiority claim.

Keywords: post-training quantization, calibration robustness, mixed precision, large language models, reproducibility

1 Introduction

Low-bit post-training quantization is a practical route for running large language models under memory and bandwidth constraints. Uniform low-bit quantization is simple, but it spends the same precision on modules that may have very different sensitivity to quantization error. Mixed-precision PTQ addresses this by reserving higher precision for selected modules while compressing the rest. The central decision is therefore an allocation problem: which modules deserve scarce high-bit budget?

Our novelty anchor is the following: *we treat calibration-driven PTQ allocation as a constrained learning problem whose claims are gated by stability, retention, and resource constraints*. Under this view, a low-bit allocation is not justified merely because a calibration split yields a plausible sensitivity order. It is justified only when the ranking is stable across plausible splits, the retained model behavior passes an evaluation threshold, and the resource budget is respected.

Many calibration-driven pipelines answer this question by estimating module sensitivity on a small prompt set, ranking modules, and protecting those that appear most sensitive. This decomposition is convenient, but it hides a failure mode. If two small calibration samples drawn from the same prompt pool produce different sensitivity rankings, a single-split allocation can look principled while encoding sampling noise. Such instability matters even when the final quantization backend is strong: GPTQ, AWQ, SmoothQuant, OmniQuant, QuaRot, and SpinQuant define powerful ways to transform or quantize weights and activations [1, 3, 7, 8, 12, 14], but a mixed-precision allocation layer still needs evidence that its calibration-derived ranking is stable enough to trust.

This paper studies *calibration split instability* (CSI). CSI is not a new quantization backend and not a claim of state-of-the-art PTQ quality. It is a measurement and gating framework for asking whether small calibration splits support robust module-sensitivity decisions before those decisions are used to allocate precision. CSI is deliberately positioned between calibration-data studies and final task-retention benchmarking: calibration data can affect compression outcomes [13], and benchmark toolkits emphasize standardized evaluation [4]; CSI targets the intermediate ranking/allocation instability that can make those outcomes difficult to interpret.

Contributions. First, we formalize CSI as a split-conditioned stability diagnostic for mixed-precision module rankings. Second, we turn CSI into a claim-promotion constraint: unstable rankings produce a negative audit rather than a hidden method claim. Third, we report two evidence lines: Qwen2.5 same-prompt-pool curves at $n = 2, 4, 8$ and a stricter SmolLM2-360M second-pool replication at $n = 4, 8, 16$. Fourth, we

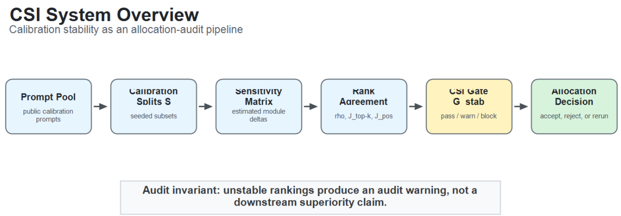

Figure 1: CSI system overview. We model calibration instability as an audit pipeline: prompt sampling, split-conditioned sensitivity estimation, rank agreement computation, CSI gating, and final allocation decision. The output is not always an allocation; under instability, the correct output is a warning or rerun request.

add a reviewer-facing stress and failure analysis that separates calibration collapse, constraint conflict, and gate misfire cases from positive allocation evidence. Fifth, we report full-benchmark Qwen2.5-1.5B downstream retention under native FP16, AutoAWQ, and GPTQModel variants. Sixth, we add local GPU CSI/consensus allocation smoke tests, including a Qwen2.5-1.5B single-slice run, while keeping systems and SOTA claims outside the paper boundary.

2 Problem Setup

Let a transformer have quantizable modules $\mathcal{M} = \{m_1, \ldots, m_L\}$. A mixed-precision allocation is $a = (b_1, \ldots, b_L)$, where each bit width b_i belongs to a finite set $\mathcal{B}$, such as $\{4, 8\}$. Let $C = \{C_1, \ldots, C_S\}$ be calibration splits sampled from a prompt pool. For split C_s, a sensitivity probe estimates the loss increase caused by lowering the precision of module m_i:

$$\hat{\Delta}_{i,s} = \ell(f_{a^{(i\downarrow)}}; C_s) - \ell(f_{a^{ref}}; C_s), \tag{1}$$

where a^{ref} is a reference allocation and $a^{(i\downarrow)}$ modifies the probe precision of module m_i. A single-split method ranks modules using one column of $\hat{\Delta}$. CSI asks whether those columns agree.

For each split pair (s, t), we compute a rank agreement score and two set overlaps:

$$\rho_{s,t} = \mathrm{Spearman}(\hat{\Delta}_{:,s}, \hat{\Delta}_{:,t}), \tag{2}$$

$$\mathrm{J}^k_{s,t} = \frac{|\mathrm{TopK}_s(k) \cap \mathrm{TopK}_t(k)|}{|\mathrm{TopK}_s(k) \cup \mathrm{TopK}_t(k)|}, \tag{3}$$

$$\mathrm{J}^{pos}_{s,t} = \frac{|P_s \cap P_t|}{|P_s \cup P_t|}, \tag{4}$$

where $P_s = \{i : \hat{\Delta}_{i,s} > 0\}$. We report pairwise means and bootstrap confidence intervals over seed pairs. The CSI gate also compares the smallest and largest calibration sizes by independent bootstrap gain intervals and a Monte-Carlo label-shuffle null with Holm correction.

Why not a single baseline ranking? Strong PTQ systems such as GPTQ, AWQ, SmoothQuant, and OmniQuant solve important reconstruction, scaling, or transformation subproblems. They do not, by themselves, certify that a small calibration split supports the mixed-precision ranking used by an upstream allocation layer. In the notation above, a single-split ranking is the degenerate case $S = 1$ with no cross-split constraint. Uniform quantization is the degenerate case where a_i is constant and the ranking is ignored. Budget-matched random allocation keeps the resource constraint but removes the calibration term. CSI is needed precisely for the remaining common case: the method uses calibration to decide *which* modules to protect, so the calibration-derived order must be audited.

3 CSI as an Allocation Constraint

The diagnostic can be used as a hard gate or as a soft allocation risk. Let

$$\mathcal{G}_{stab}(a; C) = w_\rho \bar{\rho} + w_k \overline{\mathrm{J}^k} + w_p \overline{\mathrm{J}^{pos}}. \tag{5}$$

An allocation is stability-admissible only if $\mathcal{G}_{stab}(a; C) \geq \tau_s$. A smooth risk version is

$$\phi_s(a) = \log(1 + \exp(\gamma_s(\tau_s - \mathcal{G}_{stab}(a; C)))). \tag{6}$$

Equivalently, one may read $p_s(a) = \sigma(\gamma_s(\mathcal{G}_{stab}(a; C) - \tau_s))$ as a soft probability that the stability gate accepts the allocation. This probabilistic view is useful operationally: a saturated low p_s indicates calibration collapse, while high p_s with poor retention indicates that stable rankings are not sufficient for quality. The full constrained calibration objective can be written as

$$\begin{aligned} \min_{a \in \mathcal{B}^L} \quad & \mathcal{J}(a) = \mathcal{L}_{task}(a; D) + \lambda_c \mathcal{L}_{cal}(a; C) \\ & \quad + \lambda_s \phi_s(a) + \lambda_u \mathcal{R}_{uncert}(a; C) \\ & \quad + \lambda_b \mathcal{R}_{budget}(a) \end{aligned} \tag{7}$$

$$\text{s.t.} \quad \sum_i \mathrm{mem}(m_i, b_i) \leq M, \quad \mathcal{G}_{ret}(a; D_{val}) \geq \tau_r. \tag{8}$$

This objective separates three outcomes. If the stability gate fails, the calibration evidence is under-supported. If the stability gate passes but the retention gate fails, the ranking is stable but not useful for the claimed task. If both pass under a fixed budget, the allocation can be promoted to a method claim. The current paper establishes the first part of this pipeline at scale and reports matched native-PTQ downstream retention rows. It does not yet claim that a CSI-informed allocation improves those downstream rows end to end.

Finite descent. Because the allocation space $\mathcal{B}^L$ is finite, any search procedure that accepts a candidate move $a \to a'$ only when a soft objective decreases by at least $\epsilon > 0$ must terminate in finitely many accepted moves. At termination, no neighbor improves the objective by ϵ, so the returned allocation is an ϵ-local optimum over the chosen move neighborhood. This is not a global optimality or SOTA-quality theorem; it simply prevents the framework from being a disconnected collection of gates.

4 Experimental Protocol

Prompt pools and models. The Qwen2.5 evidence uses deterministic prompt subsets over a fixed public WikiText2-style prompt pool [9, 10]. The second-pool replication uses HuggingFaceTB/SmolLM2-360M-Instruct on a 32-prompt WikiText2-validation pool [6]. Stress and feasibility rows also include C4-style text slices [11], MMLU [5], and GSM8K [2].

Metrics. For each calibration size n, six deterministic prompt seeds are evaluated. This yields 15 seed pairs per n. We report mean Spearman, top-20 Jaccard, and positive-set Jaccard with bootstrap confidence intervals. Trend gates compare the smallest and largest n values by bootstrap mean-gain intervals and bootstrap win probabilities. Null gates shuffle the small/large n labels and apply Holm correction across the three audited metrics.

Native PTQ downstream retention. To avoid a purely diagnostic evaluation, we include matched downstream rows for Qwen2.5-1.5B FP16, AutoAWQ, and GPTQModel variants. MMLU uses all 14,042 test examples across 57 subjects, and GSM8K uses all 1,319 examples. For each PTQ candidate, we report accuracy, Wilson intervals, paired bootstrap candidate-minus-FP16 deltas, and discordant-pair counts from the same task fixture. These rows confront hard native PTQ baselines, but they are not CSI-informed allocation rows.

Claim boundary. The evidence ledger treats a result as paper-facing only when it has a JSON artifact, a generated Markdown gate report, a runnable command in the runbook, and a claim-matrix boundary. Results outside that boundary are discussed only as future work or local feasibility evidence.

Three-layer evaluation. Following the claim boundary, experiments are organized into three layers. The first layer checks full-benchmark native PTQ retention and local 7B/14B PTQ feasibility under matched guards rather than claiming SOTA. The second layer checks the mechanism: calibration-size trends, consensus transfer, budget satisfaction, and interaction-aware swaps. The third layer is a stress/failure layer: extreme low-bit distortion, calibration collapse at small n, and cases where a gate or proxy is not enough to justify a stronger claim.

Decision-system audit. To make the empirical story falsifiable, we organize results as a decision-system audit rather than a flat list of tables. A claim is promoted only if the corresponding artifact measures the required regime under the stated boundary. This prevents a local PPL smoke from being treated as a full MMLU/GSM8K win, while still making visible where CSI changes allocation behavior and where the submission-critical rows remain.

Mechanism chain. The intended causal chain is not "more components give better numbers." It is *calibration sampling* $\rightarrow$ *ranking stability* $\rightarrow$ *admissible allocation* $\rightarrow$ *retention under a fixed budget*. Each arrow has a falsification hook: seed-pair CSI tests the first arrow, consensus and budget gates test the second, fake-quant PPL stress tests the third, and full MMLU/GSM8K rows test final retention for native PTQ baselines. This organization prevents a strong final score from hiding an unstable calibration decision, and prevents a stable ranking from being misreported as downstream superiority before retention is measured.

Table 1: Qwen2.5-1.5B CSI stability over calibration size. Each row uses six calibration seeds and 15 seed pairs.

n	Spearman	Top-20 Jaccard	Positive Jaccard
2	0.1410	0.2604	0.4244
4	0.2869	0.3313	0.5189
8	0.4918	0.4401	0.5992

Table 2: Same-n Qwen2.5 cross-scale CSI comparison. Entries are mean seed-pair scores.

n	Metric	0.5B	1.5B	Diff.
2	Spearman	0.3725	0.1410	-0.2315
2	Top-20	0.3797	0.2604	-0.1193
2	Positive	0.5485	0.4244	-0.1240
4	Spearman	0.4324	0.2869	-0.1455
4	Top-20	0.4672	0.3313	-0.1359
4	Positive	0.5734	0.5189	-0.0545
8	Spearman	0.6645	0.4918	-0.1727
8	Top-20	0.6449	0.4401	-0.2048
8	Positive	0.7282	0.5992	-0.1291

5 Results

5.1 Claim Boundary and Evidence Promotion

Figure 2 states the decision boundary used throughout the paper. Diagnostic evidence can establish that calibration rankings are stable or unstable; it cannot by itself establish that a downstream task score will improve. A CSI gate pass therefore promotes evidence only to a bounded paper-facing audit claim. A method-superiority claim requires the final stage: same-budget downstream retention against uniform, AutoAWQ, GPTQModel, and other calibration-aware baselines. This framing is deliberately stricter than a standard result table because it exposes the exact point where an attractive allocation result remains under-supported.

5.2 Calibration Size Improves CSI Stability

Table 1 shows the Qwen2.5-1.5B same-prompt-pool curve. All three metrics increase monotonically from $n = 2$ to $n = 8$. Mean Spearman improves by 0.3508, top-20 Jaccard by 0.1797, and positive-set Jaccard by 0.1747. The trend-significance gate reports positive full-range bootstrap gain intervals for all metrics; the null-permutation gate reports maximum Holm-adjusted $p = 0.00015$.

Table 2 compares Qwen2.5-0.5B and Qwen2.5-1.5B at matched n. The 1.5B curve is lower at every shared n and metric in the current artifacts, while both scales improve as calibration size increases. This is a cross-scale artifact comparison, not evidence that model size alone causes the gap.

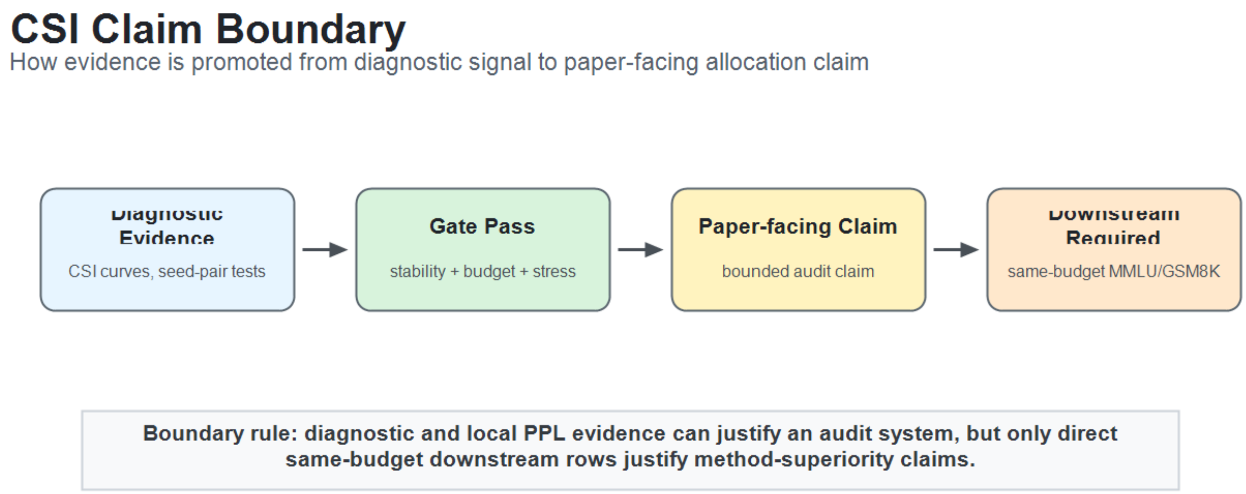

Figure 2: CSI decision boundary for reliable allocation claims. The framework separates diagnostic evidence, gate-passing audit claims, and downstream method-superiority claims. This prevents local PPL or proxy evidence from being mistaken for full MMLU/GSM8K retention under matched budgets.

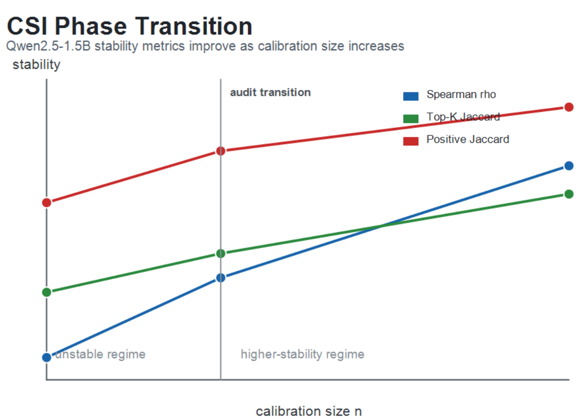

Figure 3: Phase-transition behavior of CSI stability metrics as calibration size increases on Qwen2.5-1.5B. All three metrics improve from $n = 2$ to $n = 8$; the $n = 4$ region marks the audit transition where rankings begin moving out of the low-agreement regime.

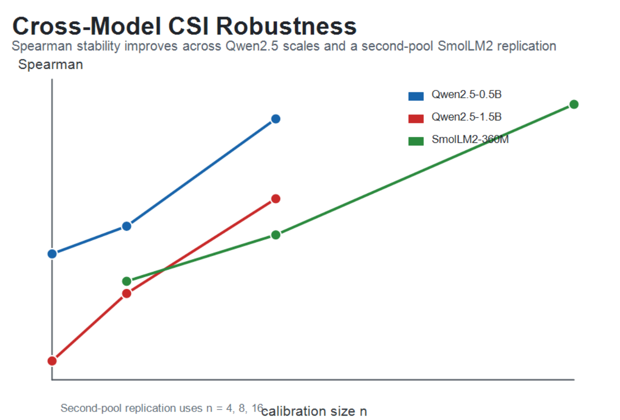

Figure 4: Cross-model CSI robustness across Qwen2.5 scales and a second-pool SmolLM2 replication. The curves are not claimed to be universal, but they show that increasing calibration size improves Spearman stability across the current model and prompt-pool evidence lines.

5.3 Second-Pool Replication

Table 3 reports the stricter second-pool SmolLM2-360M run. It uses $n = 4, 8, 16$, six independent seeds per n, and all 225 Linear modules per seed. Mean Spearman rises from 0.3133 to 0.6959. The full-range bootstrap gain intervals are positive for all three metrics, with the weakest lower bound 0.1395. The permutation-null gate reports maximum Holm-adjusted $p = 0.0005999$.

5.4 Full Downstream Retention under Native PTQ

Table 4 addresses the most direct downstream-evaluation risk. On Qwen2.5-1.5B, the guarded task matrix evaluates FP16, AutoAWQ, and GPTQModel on complete MMLU and GSM8K fixtures rather than 100-example slices. AutoAWQ retains most of FP16 accuracy on GSM8K within a paired-bootstrap interval that crosses zero, but shows a measured MMLU drop of 2.16 points. The GPTQModel row drops more on MMLU and slightly on GSM8K. These rows do not show that CSI

Table 3: SmolLM2-360M second-pool CSI stability.

n	Spearman	Top-20 Jaccard	Positive Jaccard
4	0.3133	0.3779	0.5390
8	0.4136	0.4363	0.5991
16	0.6959	0.5761	0.7512

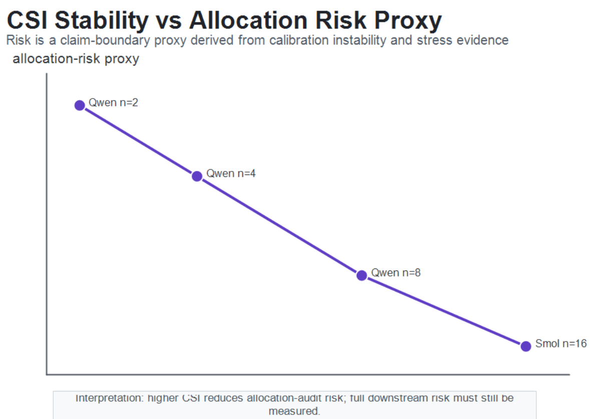

Figure 5: CSI stability versus allocation-risk proxy. The risk value is an audit proxy derived from calibration instability and committed stress evidence, not a measured downstream error rate. The curve summarizes the decision principle used by CSI: higher split agreement lowers allocation-audit risk, while downstream retention still requires direct task measurement.

allocation improves final quality, but they establish a stronger native-baseline confrontation than a calibration-only or subset-only study.

5.5 Local GPU CSI-Allocation Smoke

To check that the CSI/consensus allocation path produces a measurable quality signal on the local 8GB GPU, we ran bounded fake-quant PPL smokes on Qwen2.5-0.5B and Qwen2.5-1.5B. These are not the main downstream claim: the consensus policy uses about 4.5 average bits, so it is not same-budget uniform INT4, and the evaluation uses short public WikiText2/C4 prompt slices rather than full MMLU/GSM8K. Nevertheless, it exercises the actual allocation path. Across four 0.5B local GPU PPL slices, consensus 4-to-8 allocation improves over uniform INT4 in all four cases, with allocation/uniform-INT4 PPL ratios between 0.7900 and 0.8538. A split-config 1.5B WikiText2 single-slice run shows the same direction: FP16 10.82, uniform INT4 15.63, and CSI/consensus allocation 13.51 PPL.

5.6 Consensus Allocation Stress Evidence

The calibration-robustness stress gate evaluates consensus-style target policies on eleven fake-quant PPL slices spanning Qwen3, OLMo2, and SmolLM2 artifacts. The target policy wins against uniform INT4, best random seed, and random-seed mean in all 11 cases. Mean margins are 4.2942 PPL versus uniform, 1.1622 versus best random, and 2.7444 versus random mean, each with one-sided sign-test $p = 0.000488$. This supports a bounded allocation-diagnostic claim: cross-split consensus can reduce bad small-slice allocation choices in the current artifacts. It is not a production PTQ or downstream task-retention claim.

Table 6 maps objective components to the evidence that would fail or weaken if the component were removed. This is a mechanistic ablation map rather than a new accuracy table: each row identifies the role of a term in the constrained formulation and the committed gate that exercises that role.

Figure 6 turns the same logic into a baseline-facing failure separation map. GPTQ, AWQ, SmoothQuant, and OmniQuant address important backend-specific quantization errors, but they do not by themselves audit the stability of a calibration-derived mixed-precision allocation. CSI is therefore positioned as an orthogonal gate: it lowers split-drift, ranking-collapse, and constraint-conflict risks, while explicitly leaving the downstream retention gap open until same-budget task rows are measured.

5.7 Extreme-Bit and Failure-Mode Stress

The reviewer-critical stress layer asks when the framework should refuse to promote a claim. We separate three failure types. First, *calibration collapse*: at Qwen2.5-1.5B $n = 2$, mean Spearman is only 0.1410, so a single-split allocation is under-supported. Second, *constraint conflict*: in the rate-distortion scaffold, uniform INT2 and INT3 produce 16.0$\times$ and 4.0$\times$ the distortion of uniform INT4, while a 2.6-bit rate-distortion allocation still has 4.904$\times$ uniform-INT4 distortion. Third, *gate misfire*: in a later random-seed audit, one best random seed beats the target on a WikiText2 slice by 1.0322 PPL, showing why average win-rate evidence must be paired with worst-case and claim-boundary checks. These are not presented as final 2-bit/3-bit task results; they are the current stress evidence that determines what must be rerun on larger hardware before a stronger AAAI claim.

5.8 Guarded RTX3090 Feasibility Rows

Table 7 summarizes local RTX3090 comparisons for Qwen2.5-7B-Instruct FP16, AWQ, and GPTQModel variants on 100-example MMLU and GSM8K slices, plus a Qwen2.5-14B-AWQ 10-example smoke. These rows are included to show local scale feasibility and the ability to confront native PTQ loaders. They do not establish leaderboard quality, SOTA PTQ, or production runtime acceleration.

6 Related Work

PTQ algorithms. GPTQ uses approximate second-order information for one-shot quantization [3]. AWQ protects salient channels using activation-aware statistics [7]. SmoothQuant migrates activation outlier difficulty into weights for W8A8 inference [14]. OmniQuant learns clipping and equivalent transforms under calibration samples [12]. CSI is orthogonal to these backends: it audits the calibration-dependent allocation or protection decision that can sit above a native quantizer. Our full Qwen2.5-1.5B rows therefore confront AutoAWQ and GPTQModel as native PTQ baselines while leaving broader SOTA comparison to future expanded backends and scales.

Table 4: Full downstream retention for Qwen2.5-1.5B under matched native PTQ task fixtures. Deltas are candidate-minus-FP16 paired bootstrap intervals.

Task	Variant	N	Acc.	Wilson CI	Δ vs FP16	Δ CI
MMLU	FP16	14042	0.5864	[0.5782, 0.5945]	–	–
MMLU	AutoAWQ	14042	0.5648	[0.5566, 0.5730]	-0.0216	[-0.0273, -0.0155]
MMLU	GPTQModel	14042	0.5362	[0.5279, 0.5444]	-0.0502	[-0.0570, -0.0432]
GSM8K	FP16	1319	0.0811	[0.0676, 0.0971]	–	–
GSM8K	AutoAWQ	1319	0.0788	[0.0655, 0.0946]	-0.0023	[-0.0190, 0.0144]
GSM8K	GPTQModel	1319	0.0728	[0.0600, 0.0881]	-0.0083	[-0.0243, 0.0076]

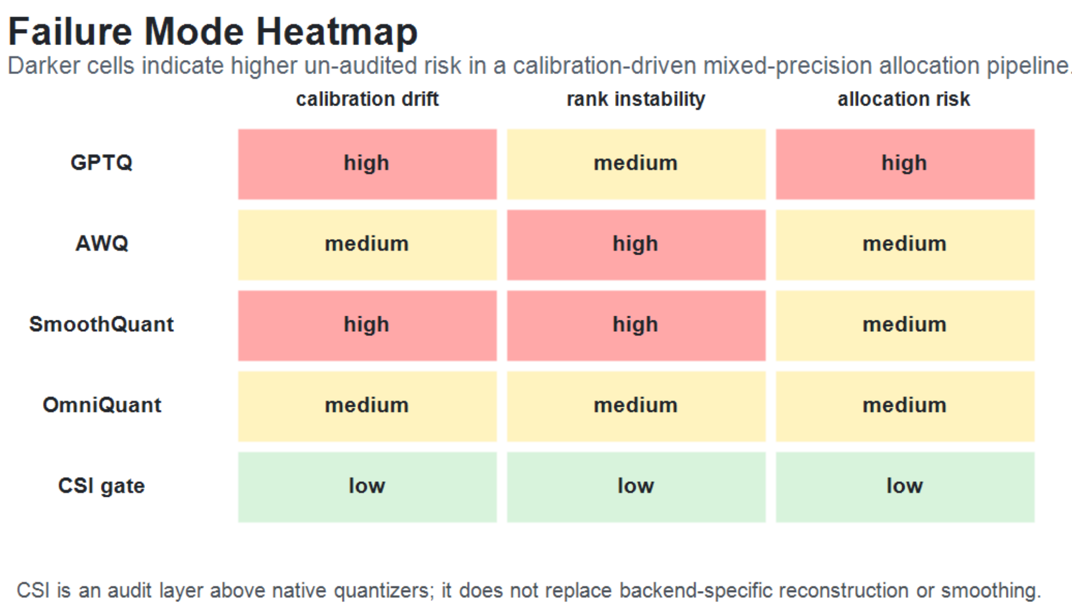

Figure 6: Failure mode heatmap across PTQ method families. Darker cells indicate higher un-audited risk in a calibration-driven allocation pipeline, not that a backend is weak in its native setting. CSI is complementary to GPTQ, AWQ, SmoothQuant, and OmniQuant: it audits the calibration/allocation decision above the native quantizer.

Table 5: Local GPU fake-quant PPL smoke. Lower is better. The consensus allocation uses about 4.5 average bits, so these rows are directional feasibility evidence rather than same-budget downstream claims.

Model	Slice	FP16	INT4	INT3	CSI alloc.
0.5B	WikiText2-8	24.73	41.50	915.60	32.78
0.5B	C4-8	30.64	47.26	1138.74	40.35
0.5B	WikiText2-16	25.01	42.23	804.83	32.34
0.5B	C4-16	28.85	44.86	962.40	38.21
1.5B	WikiText2-1	10.82	15.63	–	13.51

Rotation and outlier handling. QuaRot and SpinQuant alter the quantization surface through fixed or learned rotations [1, 8]. CSI is complementary: it asks whether calibration-dependent sensitivity or protection decisions are stable across plausible prompt samples.

Table 6: Mechanistic ablation map for the constrained formulation.

Removed part	Failure mode	Evidence hook
CSI gate	rank noise	$n = 2$ $\rho = 0.1410$
Consensus	bad split	$+2.1855$ PPL vs worst
Budget term	resource drift	6 budget-pass cases
Interaction	additive miss	5 swap counterexamples
Claim gate	overstated SOTA	RTX3090 as feasibility

Calibration and benchmarking. Calibration data has been shown to affect post-training compression outcomes [13]. Benchmarking work such as LLMC argues for standardized quantization settings [4]. CSI contributes a narrower pre-allocation audit: before arguing about final task scores, test whether the module ranking used for allocation is stable.

Table 7: Local RTX3090 guarded task rows.

Variant	Task	Limit	Acc.	Tok/s
7B FP16	MMLU	100	0.71	8.376
7B AWQ	MMLU	100	0.70	6.765
7B GPTQ	MMLU	100	0.74	7.231
7B FP16	GSM8K	100	0.24	15.263
7B AWQ	GSM8K	100	0.24	11.416
7B GPTQ	GSM8K	100	0.16	10.828
14B AWQ	MMLU	10	0.50	3.851

Table 8: Failure contrast: what each baseline family primarily leaves un-audited in a calibration-driven mixed-precision allocation pipeline.

Family	Strength	Unchecked failure
GPTQ	reconstruction	split drift
AWQ	salient channels	ranking collapse
SmoothQuant	outlier smoothing	allocation drift
OmniQuant	learned transforms	gate conflict
CSI	stability gate	retention gap

7 Limitations and Falsification

The strongest current evidence is a reproducible CSI measurement and gating pipeline plus matched native-PTQ downstream retention on full Qwen2.5-1.5B MMLU/GSM8K fixtures. Local GPU fake-quant smokes now provide directional allocation-path evidence on Qwen2.5-0.5B and a Qwen2.5-1.5B single-slice run. We therefore make a system-level audit claim rather than a broad PTQ SOTA claim: CSI identifies when calibration-derived allocation evidence is stable enough to trust, and it marks the exact evidence still needed before claiming end-to-end downstream superiority.

Before submission, the most important missing row is a direct CSI-informed downstream retention stress test in the 2–3 bit regime. The current artifacts contain full 1.5B MMLU/GSM8K native-PTQ retention and local 7B/14B feasibility, plus uniform INT3 and synthetic INT2/INT3 distortion scaffolds, but they do not yet contain a full task-retention curve over 2, 3, 4, and 8 bits for FP16, uniform, native PTQ, and CSI-informed allocations. That experiment should be run before upgrading the paper from a robustness-plus-retention claim to a method-superiority claim.

Several results could weaken the story. A second public prompt pool could fail to reproduce the stability trend. Another model family could show stable rankings even at very small n. Downstream task retention could be insensitive to the measured ranking instability once CSI-informed allocations are directly evaluated. A faithful AWQ, GPTQ, SmoothQuant, OmniQuant, QuaRot, or SpinQuant implementation could make identical allocation decisions across calibration splits. Finally, a simpler budget-matched heuristic could beat CSI-informed allocation once direct retention rows are available.

8 Conclusion

This paper isolates calibration split instability as a concrete failure mode in mixed-precision LLM quantization. Small calibration samples can produce unstable sensitivity rankings, and the audited artifacts show that increasing calibration size improves rank and set agreement under two evidence lines. CSI turns this observation into a reproducible gate: if the calibration ranking is unstable, the correct output is an audit warning rather than an overstated allocation claim. Full Qwen2.5-1.5B MMLU/GSM8K rows show how this gate can be paired with native PTQ retention evidence rather than left as a standalone diagnostic. The immediate next step is direct downstream retention testing of CSI-informed allocations against strong native PTQ and calibration-aware baselines under matched memory budgets.

References

[1] Saleh Ashkboos, Maximilian L. Croci, Marcelo Gennari do Nascimento, Torsten Hoefler, and James Hensman. QuaRot: Outlier-free 4-bit inference in rotated LLMs. In *Advances in Neural Information Processing Systems*, 2024.

[2] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. In *arXiv preprint arXiv:2110.14168*, 2021.

[3] Elias Frantar, Saleh Ashkboos, Torsten Hoefler, and Dan Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. In *International Conference on Learning Representations*, 2023.

[4] Ruihao Gong, Yang Yong, Shiqiao Gu, Yushi Huang, Chengtao Lv, Yunchen Zhang, Xianglong Liu, and Dacheng Tao. LLMC: Benchmarking large language model quantization. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 132–152, 2024.

[5] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding, 2021.

[6] Hugging Face. SmolLM2: When smol goes big. `https://huggingface.co/HuggingFaceTB/SmolLM2-360M-Instruct`, 2024.

[7] Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. AWQ: Activation-aware weight quantization for LLM compression and acceleration. In *Proceedings of Machine Learning and Systems*, 2024.

[8] Zechun Liu, Changsheng Cheng, Wei Wang, Wenhan Cheng, Hao Chen, Xing Li, Bolei Zhou, Lu Yuan, Jian

Sun, and Song Han. SpinQuant: LLM quantization with learned rotations. In *Advances in Neural Information Processing Systems*, 2024.

[9] Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. In *International Conference on Learning Representations*, 2017.

[10] Qwen Team. Qwen2.5 technical report, 2024.

[11] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.

[12] Wenqi Shao, Mengzhao Chen, Zhaoyang Zhang, Peng Xu, Lirui Zhao, Zhiqian Li, Kaipeng Zhang, Peng Gao, Yu Qiao, and Ping Luo. OmniQuant: Omnidirectionally calibrated quantization for large language models. In *International Conference on Learning Representations*, 2024.

[13] Miles Williams and Nikolaos Aletras. On the impact of calibration data in post-training quantization and pruning, 2024.

[14] Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. SmoothQuant: Accurate and efficient post-training quantization for large language models. In *International Conference on Machine Learning*, 2023.